Sir Syed University of Engineering & Technology
Faculty of Computing and Applied Sciences
Department of Computer Science & Information Technology

Mid Term Examinations (Fall 2025) Solution

Course Code with Title	**CS-475: Human Computer Interaction**	**Program**	Bachelor of Computer Science BS (CS)
Instructor(s)	**Engr. Farhana Ashraf**	**Semester**	7th
Start date & Time	Dec 02, 2025 at 03:30 PM	**End Time**	05:00 PM
Maximum Marks	30		

<u>IMPORTANT INSTRUCTIONS</u>:

Read the following Instructions carefully:

- Attempt All Questions.
- Please do mention your section on the top of the answer script.

Q.No.1	(**CLO_1**): (Cognitive Level C2, i.e., **Understanding**) (**PLO_2**: Knowledge for Solving Computing Problem) **(5+5+3 Marks)**

a) One of the problems of applying more than one of the design principles in interaction design is that trade-offs can arise between them. Therefore. 'Gulin' proposed that:

Introducing inconsistency can make it more difficult to learn an interface but in the long run can make it easier to use.

Do you agree with 'Gulin'? Why or why not? **Defend** your opinion.

Rubric:

Not attempted	Irrelevant answer	Explanation	Justification
Zero mark	1 mark	1-2.5 marks	1-2.5 marks

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Answer:

DILEMMA Usability Trade-Offs

One of the problems of applying more than one of the design principles in interaction design is that trade-offs can arise between them. For example, the more you try to constrain an interface, the less visible information becomes. The same can also happen when trying to apply a single design principle. For example, we saw how the more an interface is designed to "afford" through trying to resemble the way physical objects look, the more cluttered and difficult to use it can become. Consistency is another design principle that can be problematic to apply. As we saw earlier, trying to design an interface to be consistent with something can make it inconsistent with something else. Furthermore, sometimes inconsistent interfaces are actually easier to use than consistent interfaces. A trade-off, however, is that it can take longer to learn such an interface.

Grudin (1989) illustrates the consistency dilemma with an analogy to where knives are stored in a house. Knifes have a variety of forms—butter knives, steak knives, table knifes, fish knives. An easy place to put them all and subsequently locate them is in the top drawer by the sink. This makes it easy for everyone to find them and follows a simple consistent rule. But what about the knives that don't fit or are too sharp to put in the drawer, like carving knives and bread knives? They are placed in a wooden block. And what about the best knives kept only for special occasions? Another exception, as they are placed in the cabinet in the other room for safekeeping. And what about other knives like putty knives and paint-scraping knives used in home projects (kept in the garage) and jack knives (kept in one's pockets or backpack)? Very quickly the consistency rule begins to break down.

Grudin notes how in extending the number of places where knives are kept inconsistency is introduced, which in turn increases the time needed *to learn* where they are all stored. However, the placement of the knives in different places often makes it *easier* to find them because they are at hand for the context in which they are used and also next to the other objects used for a specific task (e.g. all the home project tools are stored together in a box in the garage). The same is true when designing interfaces: introducing inconsistency can make it more difficult to learn an interface but in the long run can make it easier to use.

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b) For a high school quiz application for blind and deaf school. **Identify** typical users and choose suitable combination of input and output devices to best support the intended interaction. Also explain major problems the input and output devices will solve. Clearly specify assumptions (if any) being made while classifying users and making recommendations.

Rubric:

Not attempted	Irrelevant Answer:	Typical users	Input Device	Output Device	Problem Solved
Zero Marks	1 Mark	2 Marks	1 Mark	1 Mark	2 Marks

Answer:

Typical User	Input Device	Output Device	Problem Solved
Blind	Mic or Brail Keyboard	Speaker, head phones or brail printer	They can't see so they can use voice or brail as an alternate.
Deaf	Normal Keyboard	Normal Monitor	They can see so normal keyboard and monitor is sufficient for them
Blind Deaf both	Brail Keyboard	Brail printer	They can't see and hey can't listen so they can use brail as an alternate.
Not Blind Not Deaf	Normal Keyboard	Normal Monitor	Not Applicable.

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c).**Express** your own mental model of a device through which user can communicate with each other.

Rubric:

Not attempted	Irrelevant answer	Mental Model
Zero mark	zero mark	3 marks

Answer:

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Q.No.2	(**CLO_2**): (Cognitive Level C2, i.e., **Understanding**) (**PLO_8**: Computing Professionalism and Society) **(5+5+5+2 Marks)**

(a) Explain why user experience (UX) should be considered when interacting with technology in safety-critical domains. Support your answer with suitable examples.

Rubric:

Not attempted	Irrelevant answer	Explanation	Example(s)
Zero mark	1 mark	1-2.5 marks	1-2.5 marks

Sample Answer:

Explanation:

User experience (UX must be prioritized in safety-critical domains because poor design can directly lead to human error, accidents, or even fatalities. Well-designed UX ensures clarity, reduces cognitive load, and supports operators in making correct decisions under pressure.

In safety-critical environments (healthcare, aviation, nuclear plants), even small mistakes can have catastrophic consequences. UX design reduces cognitive load and helps users make correct decisions quickly. Operators often work under pressure. Interfaces must be simple, predictable, and error-proof to avoid confusion during emergencies. UX focuses on how people actually interact with systems, ensuring that technology supports human strengths rather than exposing weaknesses. A clear, consistent interface reassures users that the system will behave as expected, which is vital when lives are at stake.

Examples of UX in Safety-Critical Domains:

Medical devices like infusion pumps or ventilators must have clear displays and controls. Poor UX can cause dosage errors or delayed treatment. Electronic health records (EHRs) designed with intuitive navigation reduce the risk of misdiagnosis or wrong patient data entry. E.g. Medical Errors: Confusing device interfaces can cause incorrect drug administration.

Cockpit interfaces are designed with redundancy and clarity. Pilots rely on well-structured dashboards to make split-second decisions. Poorly designed alarms or cluttered displays can overwhelm pilots, leading to accidents. E.g.: Poorly designed cockpit alerts can overwhelm pilots during emergencies.

Driver-assistance systems in cars (e.g., lane departure warnings) must provide feedback that is noticeable but not distracting. In rapid transit systems, intuitive control panels help operators respond quickly to emergencies.

Crane operators and ship command bridges rely on clear visual cues and controls. UX design ensures operators can act safely even in complex environments.

In industrial settings, unclear controls may lead to accidents or equipment damage.

Conclusion:

UX is not just about aesthetics—it's about safety, efficiency, and trust. Designing for clarity, simplicity, and predictability helps reduce human error in high-stakes environments. Examples from healthcare, aviation, and transportation show how good UX can save lives and prevent disasters. Would you like me to create a comparison table showing how UX principles differ across healthcare, aviation, and transportation safety-critical systems? It could make the distinctions even clearer.

(b) You go into the field. You have told your interviewees that their comments to you are private. A senior VP asks you "who did well in the study?" What do you say? **Defend** your opinion.

Rubric:

Not attempted	Irrelevant answer	Answer	Justification
Zero mark	1 mark	1-2.5 marks	1-2.5 marks

Answer:

- As HCI colleague we can sometimes find it difficult to preserve anonymity (and confidentiality) while at the same time providing feedback to customers on the results of field studies.
- We can't give any answer that reveals or even narrows down the identity of an interviewee, no matter who is asking - a promise is a promise.
- To insure anonymity, I will firstly put in writing the conditions of the visit or make sure that this is a prominent topic in preparations and planning. Moreover, I will say that there will be an aggregate report from several different customers, but that no individuals will be identified.

(c) You are working in a renowned organization that the creators of persuasive technology. During course of your work it was identified that the organization is exploiting their persuasive end. As HCI professional, what do you do when confronted with knowledge that a product may misinform potential users or persuade them to do something shameful? **Defend** your opinion.

Rubric:

Not attempted	Irrelevant answer	Answer	Justification
Zero mark	1 mark	1-2.5 marks	1-2.5 marks

Answer:

The design of persuasive technologies presents some ethical issues for HCI designers. Persuasive technologies must not misinform in order to achieve their persuasive end. The creators of a persuasive technology should never seek to persuade a person or persons of something they themselves would not consent to be persuaded to do. I will try to remind and convince my managers of these ethical issues, If they are not convinced then I will quit the job.

Justification: As HCI professionals we cannot deny responsibility for the products of our work. We cannot defer responsibility to the decision makers, the managers and business sponsors that define the objectives of the system. We are responsible for ensuring that ethical issues in product development are discussed openly and not hidden away in bureaucratic closets.

(d) **Identify** any two human factors that should be considered by software engineers while preparing requirement specifications.

Rubric:

Not attempted	Irrelevant answer	1st Human factor	2nd Human factor
Zero mark	Zero mark	1 mark	1 mark

Sample Answer:

- Constant stimuli
- Making assumptions